

ODP 6

Exhibit and Client Report

Team T6, The Grape.

Andrew Peene, Arabella Carlos, James Kurtis, Jillian Vondell, Liam Alcalay

Context and Problem Statement

Spotted lantern flies (SLF) are an invasive insect from East Asia that are now present in the United States. These insects can be found in and around grape vineyards in New York and have become a massive issue for those managing these vineyards. If as few as 1-2 SLF are found in a harvest sample of 1000 grams, the whole batch is considered contaminated and will be thrown out. Additionally, SLF damage grapevines by feeding on them, decreasing the lifespan of the grapevines and their overall grape yield. As there are no opportunities for manual removal of SLF after grapes are harvested off of plants, a method to remove SLF from vineyards is required to maintain vineyard health and improve grape yield.

Due to how SLF damage grapevines in vineyards and contaminate grape harvests, many removal methods have been used in the past; however, there are many issues with these past methods. Methods that involve physical contact with grape vines, such as grippers or brushes, could damage the grapes or plants, kill the SLF while still on the plants, or expose the grapes to potentially toxic quassinoids from the SLF remains. Given how much damage SLFs already cause to grapevines, clients say that a damage rate of up to 20% is acceptable. Also, current chemical pesticides are ineffective against SLF, and increasing pesticide strength poses potential risks to the grape plants. Due to the shortcomings of removal methods that make physical contact with the plant or chemical methods, a method that does not have any of these components would be the most effective solution to remove the SLF from grapevines.

Additionally, the designed removal method would be easily incorporated into existing agricultural practices to make SLF control efficient and easy for those harvesting the grapes. Currently, one main method is manual removal of SLF from vines, which is extremely inefficient. Tractors are used frequently in vineyards for vine care, such as spraying pesticides and distributing fertilizer. A method that can be used with tractors would allow for seamless integration of SLF removal into existing routines and allow for manual labor to be used in more efficient ways.

Final Prototype and Application

The device is a suction-based attachment designed to remove spotted lanternflies from grapevines. It uses a motor-driven impeller to generate airflow, creating localized suction that draws insects into a collection chamber. The device can be mounted on either harvesting equipment during the harvest season or vineyard tractors during the off-season. By relying on suction rather than physical contact, it minimizes disturbance to the grape clusters and reduces the risk of damage to the vines or fruit while enabling efficient pest removal across vineyard rows. The device is intended to be easily mountable to existing farm equipment and is designed to be better suited to the specific debris that can be expected from its intended working environment. The final prototype features a heavy-duty aluminum mesh to separate out debris such as small sticks and accidental grapes that are to be expected from usage on grape vines. Additionally, the suctioning strength is strong enough to pull SLF off of grapevines but not strong enough to remove grapes or damage the grapevines themselves, which helps maintain the health of the grapevine. The device is intended to be heavier duty, and tractor mounted, differentiating it from a standard vacuum cleaner.

Conclusion and Recommendation

Based on our testing results, the device shows promise, but it requires substantial redesign and optimization to be implemented effectively. The primary area for improvement is the suction power relative to the surface area covered. Through our testing, we found that the use of a cone to increase the

area of suction decreased suction power and reduced the efficiency of the model SLF collected. Removing the cone allowed for quicker and more effective SLF collection. Other components of the device showed better results during testing. Since the device is charged while in use, battery life is not a concern. Additionally, based on our usage throughout testing and presenting, we found no damage to the motor, blade, and housing; however, the motion of the harvester and stress through attachment points on the device may lead to damage over time beyond what we were able to test for.

Potential improvements to be made to the device primarily focus on both increasing the suction strength of the device through decreasing the nozzle size and ensuring usage over the entire height of a grape vine. Vacuum strength could be improved by calculating the ideal nozzle size based on blade dimensions and motor power and altering the geometry of the blade housing. To ensure the device is able to cover the entire area of a grape vine with a decreased nozzle size, the device could either be attached to an actuator, allowing it to actuate up and down the length of the vine, or by placing multiple devices in vertical succession. Overall, the concept is promising but not yet ready for field testing. Further development should prioritize airflow optimization and coverage strategy to achieve effective and scalable SLF removal.

The total estimated unit cost to manufacture the device is **\$102.99** (see Bill Of Materials). This price is competitive, especially because it represents a reusable mechanical removal method rather than a recurring chemical application. For example, agricultural custom-rate estimates list crop spraying at approximately **\$15–\$22 per acre**, not including the cost of the chemical pesticide itself in many cases (Cornell Cooperative Extension of Oneida County, 2024). While pesticide spraying may have a lower upfront cost for a single application, it often requires repeated treatments throughout the growing season. In contrast, our device has a higher initial cost but could reduce recurring chemical costs over time while also offering a more targeted, contactless, and non-chemical method of spotted lanternfly removal.

Testing and Results

Longevity: Conducted testing to determine device endurance over prolonged operation by continuously operating the device for 30 minutes to test battery charge and if there was any deterioration in performance over time. **Success Criteria:** The device does not drop in performance over the period. **Results:** The device lasted the 30 minutes and had no variation in its performance over the period. **Conclusion:** We expect no issues to arise regarding the longevity of the device. It has few moving parts to seize up or malfunction during operation.

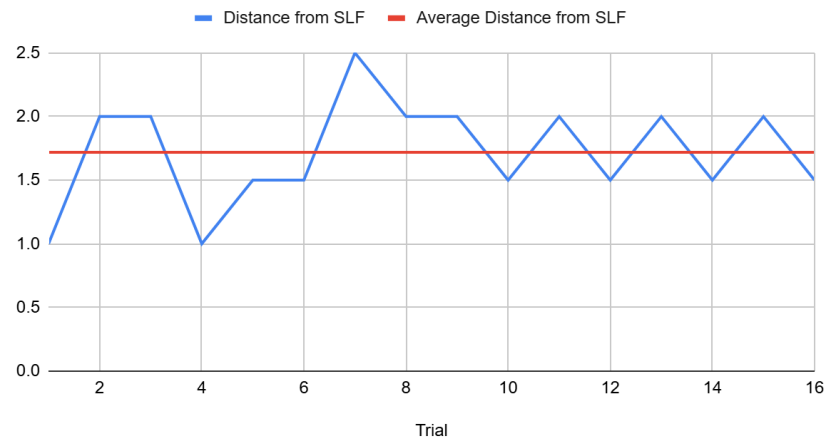
Debris Tolerance: Tested intake of various materials ranging from 1-5cm in diameter and 1-10g in mass. **Success Criteria:** the device can suck these in without damaging the fan or internal casing. **Results:** No debris caused any visible damage to the internals of the device, though many smaller and more massive debris (>5g) got trapped within the device before reaching the storage container. **Conclusion:** The device will not be damaged by any SLF or plant matter that it captures.

Mounting Load: Tested strength of mounting geometry by adding mass to the device while it supported itself. **Success Criteria:** **Results:** Successfully supported 500g of added mass (equivalent to roughly 2000 adult SLF) without any structural damage or failure. **Conclusion:** The device is more than robust enough to be mounted with a full storage of SLF.

Nozzle Distance: Tested maximum distance device could be held from the model grapevine and still collect SLF by varying the distance at which the device was held away from the model vine. The device was tested without the cone and a ruler was used to measure the distance from the end of the nozzle to the model SLF.

Success Criteria: Can collect paper without contacting the grapevine.

Distance from SLF to Suction (cm)



Results: Maximum: 2.5 cm, Minimum: 1.0 cm, Mean: 1.7185 cm, Median: 1.25 cm

Conclusion: Device in current state must be held extremely close to plants to work effectively, which likely suggests that the suction strength must be increased to ensure that the device will not make physical contact with the grape vines.

Prototype and Testing Details

We began with a mock-up to ensure that the device could be mounted on a tractor and would be geometrically possible. This mock-up left a lot to be desired and we decided that our first prototype would need to be much more space efficient.

The first prototype featured a shallow rectangular cone, a crush box in between the fan and cone, and a collection box under the crush box. The goal of this prototype was to show proof that a suction device was a viable solution. This prototype had a propeller, which we found did not generate enough suction to suck up our test SLF. We also found that the rectangular cone did not achieve our goal of widening our suction range to achieve the maximum vertical coverage of the grapevines. Additionally, we found that the intermediate crush box disrupted the airflow drastically and decreased the suction power.

The final prototype focused on improving the suction power through altering the geometry of the cone to feature a more circular design with a gradually increasing radius to create a linear airflow gradient. Additionally, to improve suction power, we replaced the propeller blade with an impeller blade with a casing attached to a motor. This was done to increase pressure differentials to increase suction power. Furthermore, to circumvent SLF or other debris getting caught in the blade and eliminate the added mass of a crushing or containment box, an aluminum mesh filter was added to direct objects through slits in the cone and mounting casing into a collection chamber. A custom mounting case was added allowing for easy addition of the device to a tractor or a harvester. The final prototype also features a charging system enabling device usage while charging, eliminating downtime between use.

Bill of Materials

Item #	Part/ Material	Description	Qty.	Supplier/Source	Cost
1	Handheld Vacuum	Functional vacuum cleaner. Includes DC motor, impeller and battery pack.	1	McMaster-Carr 7045T48	\$88.95
2	Aluminium Filter Mesh	10 x 10 Mesh Size, 0.075" Opening Size. Prevents insects and debris from entering the motor	1	McMaster-Carr 9227T57-9227T572	\$1.95 ¹
3	Housing	3D Printed PLA structure. Sections used to channel airflow into collection chamber	3	3D printed with PLA	\$11.13 ²
4	Nuts	M3 nuts. Used to attach the different housing pieces together	8	McMaster-Carr 90591A250	\$0.31
5	Bolts	M3 bolts. Used to attach the different housing pieces together	8	McMaster-Carr 91292A114	\$0.65

Total Unit Cost: **\$102.99**

¹ The aluminum filter mesh was purchased as a 2 ft × 3 ft sheet at a total cost of \$67.52. Since only a 5 in × 5 in section was required for the device, the filter mesh used represents 25 square inches of the total 864 square inches available. This results in an estimated unit cost of approximately **\$1.95** for the filter mesh.

² Additionally, the 3D-printed housing required 556.43 g of PLA. Assuming a material cost of \$0.02 per gram, the housing cost is approximately **\$11.13**.

References

California Department of Food and Agriculture. “PDCP - PD/GWSS Board - Spotted Lanternfly.” CDFA, www.cdfa.ca.gov/pdcp/board/spottedlanternfly.html. Accessed 5 May 2026.

“‘They Don’t Belong in Our Environment’: US Vineyards Battle Spotted Lanternflies as Invasive Insects Spread.” The Guardian, Guardian News and Media, www.theguardian.com/us-news/2026/may/02/invasive-lantern-flies-us-vineyards. Accessed 5 May 2026.

Cornell Cooperative Extension of Oneida County. (2024). *Custom Rates & Fees 2024*. The report lists “Spraying Crops” at **\$15–\$22/acre**.
https://s3.amazonaws.com/assets.cce.cornell.edu/attachments/65854/2024_Custom_Rates.pdf